New bounds for small two- and four-colour Ramsey numbers

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Abstract

We give upper bounds for the Ramsey numbers $\mathrm{R}(3,k)$ with $24 \le k \le 50$, for which Radziszowski's survey lists no upper bounds. They come from an exact bootstrap of lower bounds on the number of edges of triangle-free graphs with bounded independence number. It starts from the published values and lower bounds of Goedgebeur and Radziszowski. For instance, $\mathrm{R}(3,24) \le 201$, $\mathrm{R}(3,30) \le 297$ and $\mathrm{R}(3,50) \le 729$. With in addition the table of Backelin's unrefereed report, the bounds are 1 to 3 lower for $24 \le k \le 36$ except $k = 35$.

Deleting vertices from the quartic residue graphs of Lindsay and Cain gives $\mathrm{R}(4,21) \ge 260$ and $\mathrm{R}(4,24) \ge 370$. Finally, we use the four-colour form of the neighbourhood recurrence, with $\mathrm{R}(3,4,4) \le 73$ and $\mathrm{R}(4,4,4) \le 198$ from a companion paper and the survey's $\mathrm{R}(3,3,3,4) \le 149$. This gives $\mathrm{R}(3,3,4,4) \le 442$, $\mathrm{R}(3,4,4,4) \le 1521$ and $\mathrm{R}(4,4,4,4) \le 6082$. All certificates and witnesses are checked exactly by programs in the ancillary files.

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1 Introduction

For integers $s,t \ge 1$, the Ramsey number $\mathrm{R}(s,t)$ is the least n such that every red-blue colouring of the edges of K_n contains a red K_s or a blue K_t; equivalently, every graph on n vertices contains a K_s or an independent set of size t. Multicolour numbers $\mathrm{R}(t_1,\dots,t_r)$ are defined in the same way. The current state of knowledge is collected in Radziszowski's dynamic survey [14], whose revision 18 we use for all comparisons, together with the later preprint [18].

This paper is a companion to a study of three-colour numbers [5], in which local Ramsey bounds inside colour neighbourhoods are combined with double counting and with flag-algebra certificates. The present paper gives new bounds for two-colour numbers. The upper bounds are obtained by adapting the ladder procedure of [5, Section 5], and the lower bounds by deleting vertices from quartic residue graphs. It also derives consequences for four-colour numbers. Our results are listed in Table 1.

The upper bounds for $\mathrm{R}(3,k)$ (Section 2) and the lower bounds for $\mathrm{R}(4,k)$ (Section 3) are computer-assisted. Floating-point solvers and local search were used only to find certificates

number	previous	new	where
$\mathrm{R}(3,k)$, $24 \le k \le 50$	no upper bound [14]	Table 2	Theorem 2.1
$\mathrm{R}(4,21)$	≥ 242 [17]; ≥ 257 ([18] and [14, 2.3(e)])	≥ 260	Theorem 3.1
$\mathrm{R}(4,24)$	≥ 343 (implicit in [13])	≥ 370	Theorem 3.1
$\mathrm{R}(3,3,4,4)$	≤ 450 [14]	≤ 442	Theorem 4.1
$\mathrm{R}(3,4,4,4)$	≤ 1576 [14] (implicit in [3])	≤ 1521	Theorem 4.1
$\mathrm{R}(4,4,4,4)$	≤ 6306 [14]; ≤ 6301 (implicit in [3])	≤ 6082	Theorem 4.1

Table 1: Our results. For $\mathrm{R}(3,k)$ with $k \ge 24$, iterating $\mathrm{R}(3,k) \le \mathrm{R}(3,k-1)+k$, which is strict when both terms are even, or Shearer's bound [16], gives weaker explicit bounds (Table 2). Where the two rows of Table 2 differ, the smaller bound also uses the unrefereed report [2].

and witnesses, and in the negative experiments that we report. Every certificate and witness is checked in exact arithmetic by a program that does not share code with the search (Section 5).

The four-colour bounds (Section 4) follow by hand from $\mathrm{R}(3,3,3,4) \le 149$ [14] and from $\mathrm{R}(3,4,4) \le 73$ and $\mathrm{R}(4,4,4) \le 198$, which are proved in [5, Theorems 8.3 and 8.2] by certificates checked in exact arithmetic by computer.

2 Upper bounds for $\mathrm{R}(3,k)$

A $(3,k;n)$-graph is a triangle-free graph on n vertices with no independent set of size k. A $(3,k;n,e)$-graph is one with e edges, and $e(3,k,n)$ is the least such e. As in [5, Section 3], $m_k(n)$ is a *valid lower bound* if every $(3,k;n)$-graph has at least $m_k(n)$ edges.

For a vertex v of a $(3,k;n,e)$-graph G, $N(v)$ is independent, so $d_v \le k-1$. Moreover, $G_v = G - N[v]$, where $N[v] = N(v) \cup \{v\}$, is a $(3,k-1;n-1-d_v)$-graph, as an independent set of G_v extends by v. As no edge lies inside $N(v)$, G_v has $e - Z_v$ edges, where

$$Z_v = \sum_{u \in N(v)} d_u.$$

As $\sum_v Z_v = \sum_v d_v^2$, valid lower bounds m_{k-1} give

$$\sum_v \left(d_v^2 + m_{k-1}(n-1-d_v) \right) \le ne.$$

This is the inequality of Graver and Yackel [9], on which Goedgebeur and Radziszowski [7] base their bounds for $k \ge 11$. It is the two-colour analogue of the density inequality [5, Lemma 3.3]. As $ne = \sum_v nd_v/2$, it says

$$\sum_v \varphi(d_v) \le 0 \qquad \text{where} \qquad \varphi(d) = d^2 + m_{k-1}(n-1-d) - nd/2.$$

The bootstrap. We bootstrap as in the ladder of [5, Section 5], with this inequality in place of the density inequality. Level $k \ge 11$ uses the bound B_{k-1} on $\mathrm{R}(3,k-1)$ and the valid lower bounds m_{k-1} from level $k-1$. Level 11 starts from $B_{10} = 42$ [7, Theorem 6] and the values and lower bounds for $e(3,10,n)$ in [7].

Level k treats the orders $n = 0, 1, 2, \ldots$ in turn, records a valid lower bound $m_k(n)$, and stops at the first order it excludes, B_k. So $\mathrm{R}(3,k) \le B_k$. We run the procedure twice, giving two *chains*: one from the inputs of [7] alone, and one that also uses the lower bounds of [2] described after the list below. At order n the procedure is as follows.

(i) The admissible degrees are the $d \le \min(k-1, n-1)$ with $n-1-d \le B_{k-1}-1$.

(ii) An order $n \ge 1$ is excluded if $\varphi(d) > 0$ at every admissible d (in particular if there is none), as $\sum_v \varphi(d_v)$ would then be a sum of n positive terms.

(iii) Otherwise let e_0 be the largest of the following bounds. The first is the bound of Radziszowski and Kreher [7, Theorem 2]. With $K = k-1$, it gives the exact values

$$e(3,k,n) = \begin{cases} 0 & \text{for } n \le K, \\ n-K & \text{for } K < n \le 2K, \\ 3n-5K & \text{for } 2K < n \le 5K/2, \\ 5n-10K & \text{for } 5K/2 < n \le 3K, \\ 6n-13K & \text{for } 3K < n \le 13K/4-1, \end{cases}$$

and $e(3,k,n) \ge 6n-13K$ for all n. The second exists when φ is positive at the least admissible d but not at every admissible d. Let $(d', \varphi(d'))$ and $(d'', \varphi(d''))$, with $d' < d''$, be consecutive vertices of the lower convex hull of the points $(d, \varphi(d))$, d admissible, with $\varphi(d') > 0 \ge \varphi(d'')$, and let $\alpha - \beta d$ be the line through them. The second bound is $\lceil n\alpha/(2\beta) \rceil$. This line has $\beta > 0$ and lies on or below φ at every admissible d, so

$$0 \ge \sum_v \varphi(d_v) \ge n\alpha - 2\beta e.$$

(iv) The extra step below, at the orders where the search applied it (each is recorded in the certificates), may exclude e_0 and some larger values of e. The recorded $m_k(n)$ is the least $e \ge e_0$ that it does not exclude. If it excludes every e from e_0 to $\lfloor n(k-1)/2 \rfloor$, the most edges possible when every degree is at most $k-1$, then n is excluded.

In the chain with [2], the lower bounds of [2] on $e(3,k,n)$ for $11 \le k \le 13$ and $35 \le n \le 43$ are a further candidate for e_0 in (iii); at $k = 10$ they agree with those of [7]. Deleting a vertex as in [5, Lemma 3.5(a)] gives $m_k(n-1) + \max\{0, n - B_{k-1}\}$, as every degree is at least $n - B_{k-1}$. This never exceeds e_0 in either chain, so the certificates do not use it.

An extra step. We add one step, for $e = e_0, e_0+1, \ldots$ in turn. We enumerate every degree sequence with sum $2e$ and admissible entries, each at most $e - m_k(n-1)$, that satisfies the Graver–Yackel inequality. The cap on the entries holds as deleting v leaves a $(3, k; n-1)$-graph with $e - d_v$ edges.

We then try to exclude each sequence by a small linear programme. The *class* of an entry a is the set of the c_a vertices of degree a. The variables are the numbers $x_{ab} = x_{ba}$ of edges between the classes of $a \ne b$ and the numbers x_{aa} of edges inside a class. Counting edge ends, write $x_{a\to b} = x_{ab}$ for $a \ne b$ and $x_{a\to a} = 2x_{aa}$, so that $\sum_{d_v=a} Z_v = \sum_b b\, x_{a\to b}$. With $z_a = e - m_{k-1}(n-1-a)$, every $(3, k; n, e)$-graph with that degree sequence satisfies, for all entries a and b:

- $\sum_b x_{a\to b} = a\, c_a$;

- $\sum_b b\, x_{a\to b} \le c_a z_a$, as G_v has $e - Z_v \ge m_{k-1}(n-1-a)$ edges;
- $\sum_b b\, x_{a\to b} \ge c_a\big(e - \lfloor (n-1-a)(k-2)/2 \rfloor\big)$, as G_v is a $(3, k-1)$-graph and so has maximum degree at most $k-2$;
- $x_{ab} \le c_a c_b$ for $a \ne b$;
- $\sum x_{ab} \le \lfloor N_S^2/4 \rfloor$, summed over $a \le b$ in S, for every set S of classes with N_S vertices in all (Mantel's theorem);
- $\sum_{b' \le b} x_{a\to b'} \ge t\, c_a$, with $t = t(a,b)$ as follows.

For the last constraint, let q be the least entry above b, and let v be a vertex of degree a. Remove from the sequence one term a, that of v. For $0 \le j \le a$, let $L(j)$ be the sum of the j smallest remaining terms at most b, counted with multiplicity, plus $(a-j)q$. Put $L(j) = \infty$ if fewer than j such terms remain, or if no entry of the sequence exceeds b (so that q is undefined) and $j < a$. If v has exactly j neighbours of degree at most b, then $L(j) \le Z_v \le z_a$ (so no such v exists when $L(j) = \infty$). So v has at least t such neighbours, where t is the least j with $L(j) \le z_a$, and summing over the class of a gives the last constraint. If there is no such j, no vertex of degree a exists and the sequence is excluded by this *neighbour-count test*.

If every sequence is excluded, so is e, and the step moves on to $e+1$. Each exclusion is certified either by the neighbour-count test or by rational Farkas multipliers, which combine these constraints, with $x \ge 0$, into an inequality $0 \ge c$ with $c > 0$.

Theorem 2.1 (computer-assisted)**.** *From the published values and lower bounds of [7, Theorem 2 and Tables 3–5] and* $\mathrm{R}(3,10) \le 42$ *[7, Theorem 6],* $\mathrm{R}(3,k)$ *is at most the value in the row "[7] only" of Table 2 for* $24 \le k \le 50$*. With in addition the lower bounds on* $e(3,k,n)$ *in the table of [2, Section 5], it is at most the value in the row "with [2]".*

k	24	25	26	27	28	29	30	31	32
[7] only	201	216	231	247	264	280	297	315	333
with [2]	200	215	230	245	261	278	295	312	331
k	33	34	35	36	37	38	39	40	41
[7] only	351	370	389	409	429	450	471	492	513
with [2]	349	369	389	408	429	450	471	492	513
k	42	43	44	45	46	47	48	49	50
both	536	559	582	606	630	654	679	704	729

Table 2: Upper bounds for $\mathrm{R}(3,k)$ from Theorem 2.1. The two rows coincide for $k = 35$ and $k \ge 37$, and for $k \ge 42$ they are shown as one, "both". The survey [14] lists no upper bounds for these k. For comparison, $\mathrm{R}(3,k) \le \mathrm{R}(3,k-1) + k$, strict when both terms are even, iterated from the survey's $\mathrm{R}(3,23) \le 185$, gives 209, 234, 259, 371 and 1171 for $k = 24, 25, 26, 30, 50$. Shearer's bound [16] $\mathrm{R}(3,k) \le \lfloor (k-1)/f(k-1) \rfloor + 1$, with $f(d) = (d\ln d - d + 1)/(d-1)^2$, gives 223 for $k = 24$ and 792 for $k = 50$.

The two chains. Above level 10, each chain uses only the bounds B_{k-1} that it proves itself, never the unpublished values of [14]. Neither chain needs the bound $\mathrm{R}(3,10) \le 41$ of Angeltveit [1]. The report [2] is a preprint that gives little of its proofs, so the second row

rests on inputs without published proofs. The first row is 1 to 3 larger for $24 \le k \le 34$ and $k = 36$, and equals the second for $k = 35$ and $k \ge 37$.

Without the extra step, the chain from [7] alone gives the same bounds for $11 \le k \le 50$, except $\mathrm{R}(3, 20) \le 146$ and $\mathrm{R}(3, 35) \le 390$. This run is not among the certificates in the ancillary files. The chain from [7] alone also raises seven lower bounds of [7, Tables 9–11] by one, to

$$e(3, 13, 58) \ge 261, \qquad e(3, 14, 69) \ge 366, \qquad e(3, 14, 70) \ge 382,$$
$$e(3, 15, n) \ge 516,\ 534,\ 553 \text{ and } 573 \quad \text{for } n = 82, \ldots, 85.$$

Backelin's unpublished tables may already contain these.

Checking the certificates. The certificates are checked by a program that does not import the generating code. It enumerates the degree sequences itself, rebuilds each constraint from its definition, and verifies the multipliers in exact arithmetic.

- The chain with [2] has one row for each level $11 \le k \le 50$ and each order $0 \le n < B_k$, 13,376 rows in all. Each row records $m_k(n)$ and the bound of (iii) that gives e_0: the Radziszowski–Kreher bound, a value of [2], or the two ends of the hull edge. At the orders where the extra step is used, it also records a certificate for every degree sequence at each excluded e. A further row per level, not counted here, certifies the exclusion of B_k: by (ii) at every level except $k = 16$, where the extra step excludes every e at $n = 97$. The chain excludes 765 degree sequences at 53 triples (k, n, e) (including those at $k = 16$, $n = 97$), 686 by the neighbour-count test and 79 by Farkas multipliers.
- The chain from [7] alone has 13,403 rows and excludes 766 sequences at 50 triples, 684 and 82 respectively. All 40 exclusions of B_k in it are by (ii).

Comparison with the survey. For $11 \le k \le 23$ the two chains give the following bounds B_k, against the upper bounds that [14] lists.

k	11	12	13	14	15	16	17	18	19	20	21	22	23
survey [14]	50	59	68	77	87	97	109	120	132	145	157	171	185
[7] only	50	59	68	77	87	98	109	121	133	145	159	173	187
with [2]	50	59	68	77	87	97	109	121	132	145	158	172	185

So for $k \le 23$ the procedure improves no value of [14]. From [7] alone it gives $\mathrm{R}(3, 17) \le 109$ and $\mathrm{R}(3, 20) \le 145$, which [14] credits to an unpublished manuscript of Backelin and to Lesser's thesis [11], respectively. With the table of [2] it gives $\mathrm{R}(3, 16) \le 97$ and $\mathrm{R}(3, 19) \le 132$, the values that [14] lists, citing [2].

With the table of [2] it also gives $\mathrm{R}(3, 23) \le 185$. The survey [14] credits this bound to a personal communication of Backelin; it now has a derivation that can be checked, given the inputs of [2]. At $k = 18$, 21 and 22 the values of [14] come from unpublished sources, and the chain with [2] misses each by one.

For $k = 18$, both chains give $m_{18}(120) = 1020 = 120 \cdot 17/2$ from the hull. As every degree is at most 17, a $(3, 18; 120)$-graph would have exactly 1020 edges and be 17-regular. Then every G_v would be a $(3, 17; 102, 731)$-graph, as $120 - 1 - 17 = 102$ and $1020 - 17^2 = 731$.

So $e(3,17,102) \geq 732$, one more than our $m_{17}(102) = 731$, would exclude $n = 120$ and give $\mathrm{R}(3,18) \leq 120$.

3 Lower bounds for $\mathrm{R}(4,k)$

For a prime $p \equiv 1 \pmod 8$ let Q_p be the circulant graph on $\mathbb{Z}_p$ in which $x \sim y$ if and only if $x - y$ is a nonzero fourth power. The relation is symmetric, as -1 is a fourth power. Lindsay and Cain [13, Proposition 2.1] computed with cliquer that Q_{313} and Q_{457} have clique number 3 and independence numbers

$$\alpha(Q_{313}) = 21 \qquad \text{and} \qquad \alpha(Q_{457}) = 24.$$

The ancillary checks confirm $\alpha(Q_{313}) = 21$ and show that Q_{457} has an independent 24-set.

Theorem 3.1 does not use the upper bounds on $\alpha(Q_p)$. An independent set of size at least k contains one of size k. So if a vertex set S meets every independent k-set of Q_p, then $Q_p - S$ is K_4-free and has no independent k-set. Step 3 below checks this for $(p,k) = (313,21)$ and $(457,24)$, and the K_4-freeness of both graphs $Q_p - S$ is also checked directly.

Theorem 3.1 (witnesses checked by computer)**.** $\mathrm{R}(4,21) \geq 260$ *and* $\mathrm{R}(4,24) \geq 370$.

Proof. Step 1: symmetries and cosets. The maps $x \mapsto ax + b$ with a a nonzero fourth power are automorphisms of Q_p. Let O_1, O_2, O_3 be the cosets of the group of nonzero fourth powers in $\mathbb{Z}_p^*$ other than the group itself, in a fixed order, and let r_j be the least element of O_j.

Step 2: representatives. Let $k \geq 2$, let J be an independent k-set and let $u \in J$. The translate $J - u = \{x - u : x \in J\}$ is an independent k-set that contains 0, so its other elements are not fourth powers and lie in $O_1 \cup O_2 \cup O_3$. Let j be least with $J - u$ meeting O_j, say at w. Then r_j/w is a fourth power, and multiplying by it maps $J - u$ to an independent k-set that contains 0 and r_j and still avoids $O_1, \ldots, O_{j-1}$, as multiplication by a fourth power fixes each coset. Call an independent k-set a *representative* if, for some j, it contains 0 and r_j and avoids $O_1, \ldots, O_{j-1}$. So every independent k-set is the image of a representative under one of the maps above.

Step 3: the computation. The sets S, of 54 and 88 vertices, were found by a weighted local search and are listed in the ancillary files. An exact program lists all representatives for $(p,k) = (313,21)$ and $(457,24)$, by an exhaustive search for independent $(k-2)$-sets in the common non-neighbourhood of 0 and r_j outside $O_1, \ldots, O_{j-1}$. It then checks that S meets every image of every representative. So $Q_{313} - S$ is a K_4-free graph on 259 vertices with no independent 21-set, and $Q_{457} - S$ one on 369 vertices with no independent 24-set. □

Verification. Both graphs are rebuilt from the lists of deleted vertices by programs that share no code with the search. The 259-vertex graph is also checked by a direct exhaustive search for K_4 and for independent 21-sets, and the 369-vertex graph by one for K_4 and for independent 24-sets. Neither search uses symmetry, so each is independent of the orbit argument above. □

Comparison with earlier bounds. The previous bounds are as follows.

- For $\mathrm{R}(4,21)$, the survey has 242 [17], and $\mathrm{R}(4,20) \geq 252$ from a recent preprint [18] gives 257 by the inequality $\mathrm{R}(4,l) \geq \mathrm{R}(4,l-1)+5$ [14, 2.3(e)].
- For $\mathrm{R}(4,24)$, the non-neighbourhood of a vertex of Q_{457} (Q_{457} is 114-regular) has 342 vertices, is K_4-free and has no independent 24-set, as such a set would extend by the vertex. This gives $\mathrm{R}(4,24) \geq 343$. The survey lists no value for $\mathrm{R}(4,24)$. Its rules give 324, by applying this inequality twice to $\mathrm{R}(4,22) \geq 314$ [13].

The method does not reach $\mathrm{R}(4,23) \geq 320$, which would improve the bound 319 that this inequality gives from $\mathrm{R}(4,22) \geq 314$. That needs a set of at most $457-319=138$ vertices meeting every independent 23-set of Q_{457}, and the smallest we found has 147.

4 Four colours

For $\mathbf{t} = (t_1,\dots,t_4)$ and the unit vectors $\mathbf{e}_i$, the four-colour form of the neighbourhood recurrence [14, 6.1(a)] is

$$\mathrm{R}(\mathbf{t}) \leq \sum_{i=1}^{4} \mathrm{R}(\mathbf{t}-\mathbf{e}_i) - 2.$$

Indeed, consider a colouring of K_n with no colour-j K_{t_j} for any j. The colour-i neighbourhood $N_i(v)$ of a vertex v, whose size is the colour-i degree of v, contains no colour-i K_{t_i-1}, which would extend by v, and no colour-j K_{t_j} for $j \neq i$. So it has at most $\mathrm{R}(\mathbf{t}-\mathbf{e}_i)-1$ vertices. Summing over i gives

$$n-1 \leq \sum_i (\mathrm{R}(\mathbf{t}-\mathbf{e}_i)-1).$$

The recurrence improves to $\mathrm{R}(\mathbf{t}) \leq \sum_i \mathrm{R}(\mathbf{t}-\mathbf{e}_i) - 3$ when $\sum_i \mathrm{R}(\mathbf{t}-\mathbf{e}_i)$ is even and some $\mathrm{R}(\mathbf{t}-\mathbf{e}_i)$ is even. Indeed, take a colouring as above of K_n with $n = \sum_i \mathrm{R}(\mathbf{t}-\mathbf{e}_i) - 3$, which is odd. Then the display holds with equality, so every vertex has colour-i degree exactly $\mathrm{R}(\mathbf{t}-\mathbf{e}_i)-1$ for each i. For an i with $\mathrm{R}(\mathbf{t}-\mathbf{e}_i)$ even this degree is odd, and a graph on an odd number of vertices cannot have all its degrees odd.

Both statements hold with the same proofs when each $\mathrm{R}(\mathbf{t}-\mathbf{e}_i)$ is replaced by a proven upper bound for it. An entry equal to 2 may be dropped, leaving a number with fewer colours, as a colouring with no colour-i K_2 has no colour-i edges.

Theorem 4.1 (from computer-assisted inputs)**.** $\mathrm{R}(3,3,4,4) \leq 442$, $\mathrm{R}(3,4,4,4) \leq 1521$ *and* $\mathrm{R}(4,4,4,4) \leq 6082$.

Proof. With $\mathrm{R}(3,4,4) \leq 73$ [5, Theorem 8.3] and $\mathrm{R}(3,3,3,4) \leq 149$ [14], the recurrence gives

$$\mathrm{R}(3,3,4,4) \leq 2\cdot 73 + 2\cdot 149 - 2 = 442.$$

Parity does not apply, as every term is odd.

With $\mathrm{R}(4,4,4) \leq 198$ [5, Theorem 8.2] it then gives

$$\mathrm{R}(3,4,4,4) \leq 198 + 3\cdot 442 - 2 = 1522.$$

The sum $198 + 3 \cdot 442 = 1524$ is even, as are all four terms, so parity gives $\mathrm{R}(3,4,4,4) \leq 1524 - 3 = 1521$.

Finally,

$$\mathrm{R}(4,4,4,4) \leq 4 \cdot 1521 - 2 = 6082,$$

and parity does not apply, as 1521 is odd. □

The inputs and earlier bounds. The two inputs from [5] are computer-assisted. The survey's $\mathrm{R}(3,3,3,4) \leq 149$ is itself the recurrence with parity, from $\mathrm{R}(3,3,4) = 30$ [4] and $\mathrm{R}(3,3,3,3) \leq 62$ [6]: as $3 \cdot 30 + 62 = 152$ and 30 are even, it gives $152 - 3 = 149$. From the survey's $\mathrm{R}(3,4,4,4) \leq 1576$, the recurrence with parity gives $\mathrm{R}(4,4,4,4) \leq 4 \cdot 1576 - 3 = 6301$, as $4 \cdot 1576$ and all four terms are even. The survey lists 6306 and notes that 6301 is implicit in [3].

With the survey's $\mathrm{R}(3,3,4,4) \leq 450$ in place of 442, that is, without $\mathrm{R}(3,4,4) \leq 73$, the bound $\mathrm{R}(4,4,4) \leq 198$ alone gives $\mathrm{R}(3,4,4,4) \leq 1545$ and $\mathrm{R}(4,4,4,4) \leq 6178$. The survey's $\mathrm{R}(3,4,4,4) \leq 1576$, which it marks as implicit in [3], comes from Boza's analogue of the parity clause [3, Theorem 2.1], which counts the triangles of one colour and needs conditions modulo 3. With our bounds it gives nothing more: its first condition, that the right-hand side of the recurrence is not $\equiv 1 \pmod 3$, fails at each step, as 442, 1522 and 6082 are all $\equiv 1 \pmod 3$.

Four triangle-free colours. We also tried to improve $\mathrm{R}(3,3,3,3) \leq 62$ [6], that is, to exclude colourings of K_{61} with four triangle-free colours, by certificates analogous to those for $\mathrm{R}(4,4,4)$ in [5, Sections 7 and 8]. Such a certificate exists only if a flag relaxation, here with four- or five-vertex patterns and with four-colour analogues of the local caps of [5, Lemmas 2.1 and 2.3] among its constraints, is infeasible. As in Razborov's flag algebras [15], the unknowns of the relaxation are the densities of the colourings of K_4 or K_5, the patterns, in a putative colouring of K_{61}. They are subject to linear constraints from the caps, such as $|N_i(v) \cap N_j(u)| \leq \mathrm{R}(3,3) - 1 = 5$ for $u \in N_i(v)$ and $j \neq i$, and to positive-semidefiniteness constraints. A certificate is a dual solution that proves the relaxation infeasible. For comparison, the six-vertex computation of Lidický and Pfender [12, Appendix A], whose relaxation has no caps, gives only $\mathrm{R}(3,3,3,3) \leq 65$.

At $n = 61$ each colour-i degree is at most $\mathrm{R}(3,3,3) - 1 = 16$, and the four colour degrees of a vertex sum to 60. So the *profile* of a vertex, its sorted quadruple of colour degrees, is one of

$$(12,16,16,16), \quad (13,15,16,16), \quad (14,14,16,16), \quad (14,15,15,16), \quad (15,15,15,15).$$

Both relaxations are numerically feasible. Floating-point solvers found an assignment of pattern densities meeting all their constraints. Splitting by profile does not help: they found one even when every vertex is required to have profile $(12,16,16,16)$. The four-vertex relaxation was solved as a linear programme by HiGHS [10], with its semidefinite constraint imposed by eigenvector cuts until no eigenvalue was below -10^{-8}. The five-vertex relaxation was solved as a semidefinite programme by Clarabel [8] at tolerance 10^{-8}. So we obtained no such certificate from these relaxations, and no bound follows.

5 Verification, literature and limitations

Ancillary files. This paper and [5] have the same ancillary files, which are also in the directory `ancillary/` of the repository

`https://github.com/Lachy-Dauth/ramsey-neighbourhood-counting`.

They contain the programs and data, among them:

- `two_colour_r3k/` for Theorem 2.1, with the inputs from [7] and [2] and the certificates of both chains;
- `two_colour_r4k/` for Theorem 3.1, with the lists of deleted vertices, `build_power_sub.py`, which rebuilds the two graphs from them, the exact checkers `rcheck.c` (direct search) and `qpsym.c` (orbit argument), and `verify_all.sh`, which checks the SHA-256 hashes of the two graphs and runs the checkers;
- `flag5_r444/` and `flag5_r34k/`, which check the inputs $\mathrm{R}(4,4,4) \le 198$ and $\mathrm{R}(3,4,4) \le 73$ of Theorem 4.1 from [5].

The command `sh run_all.sh` runs the quick checks of both papers in a few minutes. The command `sh run_all.sh --full` adds the long ones, among them the checks of these two inputs and the direct search of the 369-vertex graph for independent 24-sets.

The checkers for this paper, in `two_colour_r3k/` and `two_colour_r4k/`, are written in Python and C with their standard libraries, and the C programs use POSIX threads. To run only these checks, use the commands in `two_colour_r3k/README.md` and the script `two_colour_r4k/verify_all.sh`. The script `run_all.sh` also needs a C++17 compiler, for the checks of [5], and uses NumPy, when it is installed, for one optional second checker in `flag5_r444/` [5, Section 10]. Negative controls are all rejected: a weakened input, a zeroed multiplier, a removed sequence certificate, a value raised by one, a claimed bound below the proved one such as $\mathrm{R}(3,24) \le 199$, and a deleted vertex put back.

Literature. We searched arXiv and the web up to 23 October 2026 for earlier statements of our new bounds, and found none.

- For $\mathrm{R}(3,k)$ with $k \ge 24$ the survey lists no upper bounds, and the upper bounds of [7] and of Lesser's thesis [11] stop at $k = 16$ and $k = 21$. Backelin's unpublished manuscript, to which [14] credits $\mathrm{R}(3,17) \le 109$, may contain bounds of this kind.
- For $\mathrm{R}(4,21)$ the best previous bound we know of, 257, follows from $\mathrm{R}(4,20) \ge 252$ in the preprint [18], which we have not checked. Without it the bound is 242 [17].
- For $\mathrm{R}(4,24)$ the survey lists nothing, and 343 is implicit in [13].
- For the four-colour numbers the previous bounds are those of [14] and [3] in Table 1.

Such a search cannot establish priority, and we would welcome pointers to any earlier source.

Limitations. For $k \le 23$ the bootstrap of Section 2 improves none of the bounds in [14]. At $k = 18$ it would give $\mathrm{R}(3,18) \le 120$ if its own lower bound $e(3,17,102) \ge 731$ were one larger. The second row of Table 2 rests on the lower bounds of [2], whose proofs are largely unpublished.

The deletion method of Section 3 does not reach $\mathrm{R}(4,23) \geq 320$. The four-colour bounds rest on $\mathrm{R}(3,4,4) \leq 73$ and $\mathrm{R}(4,4,4) \leq 198$ [5, Theorems 8.3 and 8.2], which are computer-assisted, and on $\mathrm{R}(3,3,3,4) \leq 149$, which [14, 6.1(a)] derives from $\mathrm{R}(3,3,4) = 30$ [4] and $\mathrm{R}(3,3,3,3) \leq 62$ [6], both computer-assisted.

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